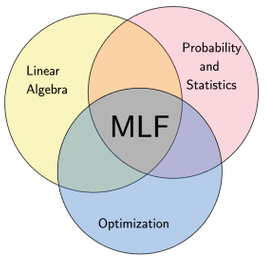
Week-9, Session-1



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Disclaimer: These notes were generated as a part of this live session. It can't be used as a standalone resource or a set of notes for this week.

1. Assumption

All functions that we consider in the optimization part are differentiable and have continuous (partial) derivatives up to order 2.

$$f''(x) \to \text{continuous}$$

$$\frac{\partial^2 f}{\partial x \partial y}, \frac{\partial^2 f}{\partial x^2}, \frac{\partial^2 f}{\partial y^2} \to \text{continuous}$$

Such functions are called **smooth functions**.

2. Unconstrained Optimization

For some function $f : \mathbb{R} \to \mathbb{R}$,

$$\min_x \quad f(x)$$

The feasible set is $\mathbb{R}$.

For some function $f : \mathbb{R}^2 \to \mathbb{R}$:

$$\min_{x,y} \quad f(x, y)$$

The feasible set is $\mathbb{R}^2$.

Terminology:

- f is called the **objective function**.
- **Feasible set** is the set of all candidates that are potential solutions to the optimization problem. For an unconstrained optimization problem, the feasible set is $\mathbb{R}^n$

3. Critical Points

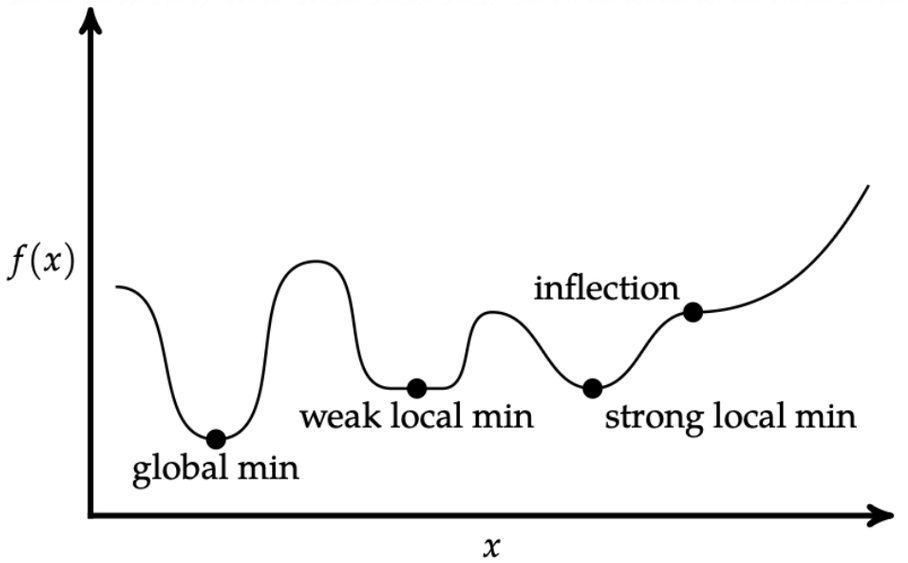
Single variable

A point x^* for which $f'\left(x^*\right) = 0$ is called a critical point of f. A more appropriate term would be stationary point. Critical points can be further classified based on the behavior of the function in the close vicinity of the point. We have the following broad categories:

- minimum
- maximum
- inflection point

Within the minimum basket, we have:

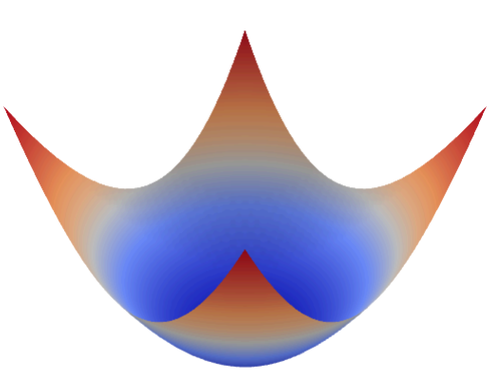
- local minimum
  - weak
  - strong
- global minimum



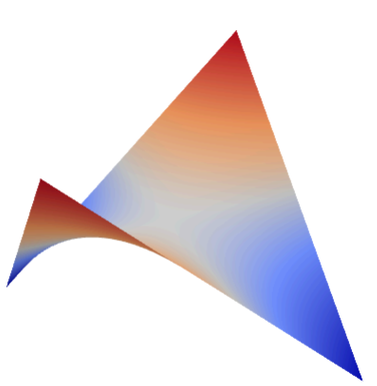
Two variables

A point (x^*, y^*) is a critical point of f if $\nabla f(x^*, y^*) = (0, 0)$. We once again have three categories:

- minimum
- maximum
- saddle point



local minimum



saddle point

A saddle point is a critical point for which in some directions the function's value increases and in others is decreases. This is shaped like a horse's saddle.

4. Understanding Necessary and Sufficient Conditions

We often come across the terms necessary and sufficient conditions in the study of optimization. It is important to understand what exactly is meant

by these two terms. Consider two statements, P and Q.

- Q is a necessary condition for P if $P \implies Q$
- P is a sufficient condition for Q if $P \implies Q$

$P \implies Q$, is shorthand for "whenever P is Q true, Q is also true" or "if P is true, then Q is also true". To make this notion clear, consider two statements P and Q:

- P: It rains.
- Q: I carry an umbrella.

$P \implies Q$ translates to "If it rains, I carry an umbrella". We say that "carry an umbrella" is a necessary condition for "it rains". In simple English, if it rains, then I necessarily carry an umbrella with me. However, note that it says nothing about what is happening when I carry an umbrella. I might carry an umbrella for other reasons. Perhaps I expect it might rain later in the day. Or it is too hot outside and I don't want to expose myself to the burning sun.

Let us now turn to the sufficiency part. When we have $P \implies Q$, P is a sufficient condition for Q. Going back to our example, "it rains" is a sufficient condition for "I carry an umbrella". In simple English, the fact that it is raining outside is sufficient for me to carry an umbrella. Once again, note that it tell us nothing about all the occasions when I might carry an umbrella. It just tell us that "raining" is one of the occasions which forces me to carry an umbrella. We don't know if it is the only occasion.

5. Necessary Conditions

Now that we have understood what necessary and sufficient conditions are, let us see what they mean in optimization problem.

5.1. First order

Single variable

If x^* is a local minimum of f, then $f'(x^*) = 0$.

We say that $f'(x^*) = 0$ is a necessary condition for x^* to be a minimum. Alternatively, if x^* is a minimum, then it necessarily means that $f'(x^*) = 0$. This is a first order condition since it deals with the first derivative.

Two variables

If x^* is a local minimum of f, then $\nabla f(x^*) = 0$.

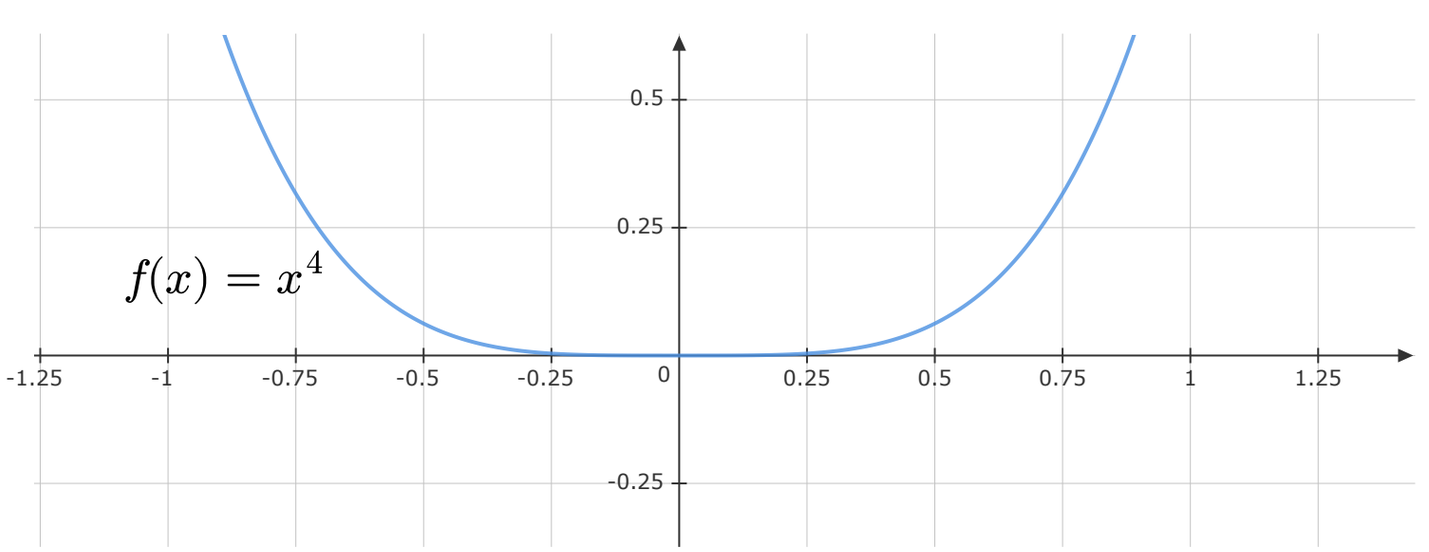
This is a straightforward generalization to two and more variables.

5.2. Second order

Single variable

If x^* is a local minimum of f, then $f''(x^*) \geqslant 0$.

This is a second order necessary condition since it involves the second derivative. The second derivative gives us information on the curvature at a point. Can you think of a function where x^* is a minimum and $f''(x^*) = 0$?



Two variables

If x^* is a local minimum of f, then $H = \nabla^2 f(x^*)$ is positive-semidefinite.

The second order necessary condition for two and more variables involves the Hessian, the matrix of second order partial derivatives. This matrix is symmetric since we have assumed that the second order derivatives are continuous. This is necessary for the mixed partial derivatives to be equal.

Caution

The necessary conditions allow us to gather a set of potential candidates that could be the minima. Every local minimum for a function has to satisfy $\nabla f(x^*) = 0$. However, not every point that satisfies $\nabla f(x^*) = 0$ qualifies as a local minimum. Hence, necessary conditions by themselves are not sufficient to find the local minima.

6. Second Order Sufficient conditions

Single variable

If $f'(x^*) = 0$ and $f''(x^*) > 0$, then x^* is a strict local minimum of f.

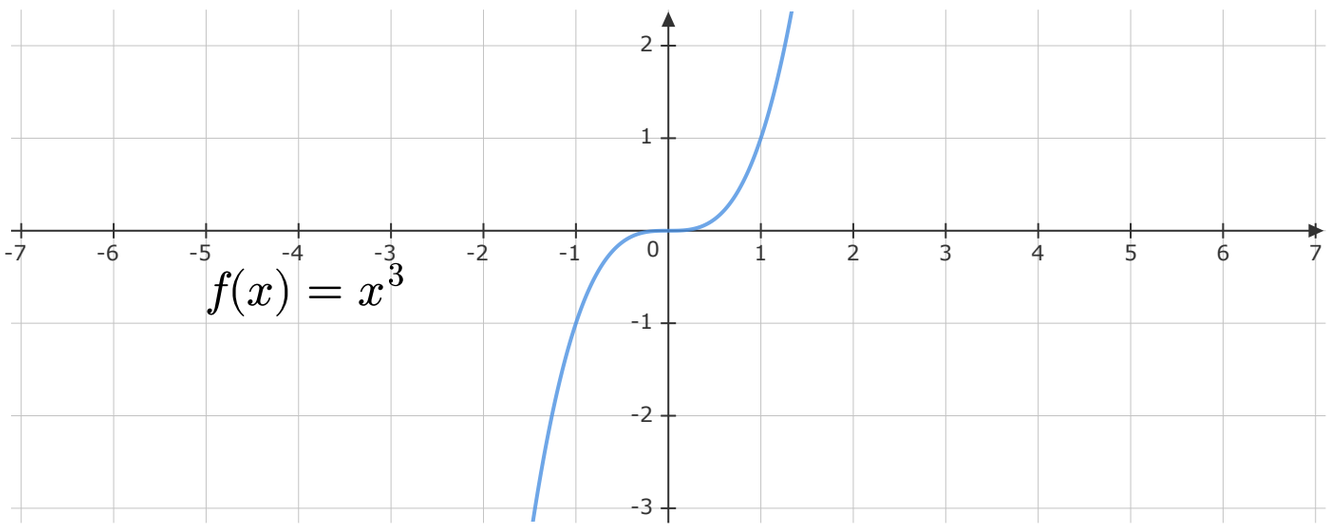
This often goes by the name of the second order test.

Two variables

If $\nabla f(x^*) = 0$ and $H = \nabla^2 f(x^*)$ is positive definite, then x^* is a strict local minimum of f.

7. Caution: First Order and Second Order Tests

While the first order and second order tests are extremely useful, they may still be inconclusive. For example, consider $\min_x \; x^3$.

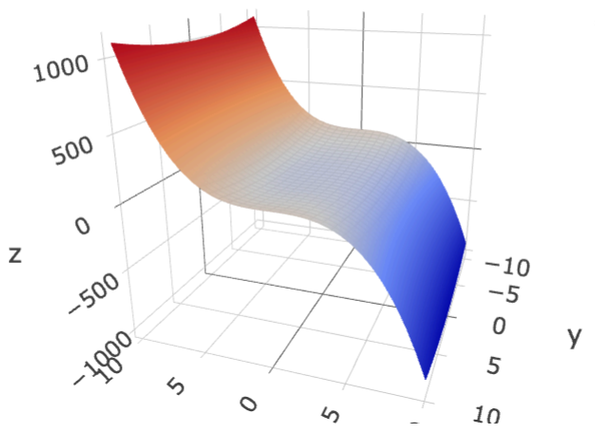


The tests give us:

- First order necessary test
  - $f'(x) = 0 \Longrightarrow x = 0$
  - If there is a minimum, it has to be at $x = 0$.
- Second order sufficient test
  - $f''(0) = 0$
  - The test is inconclusive

So the tests don't tell us anything conclusive. In this case, it is easy to visualize $f(x) = x^3$. But this isn't always the case. As another example, consider this function of two variables.

$$f(x, y) = x^3 + y^2$$



We get:

- First order
  - $(0, 0)$ is a critical point
- Second order
  - The Hessian is the zero matrix.
  - Test is inconclusive

8. Constrained Optimization with Equality Constraints

Let us now look at a constrained optimization problem with a single equality constraint.

$$\min_{x,y} \; f(x, y)$$

$$g(x, y) = 0$$

Let us look at a few examples and visualize what is happening.

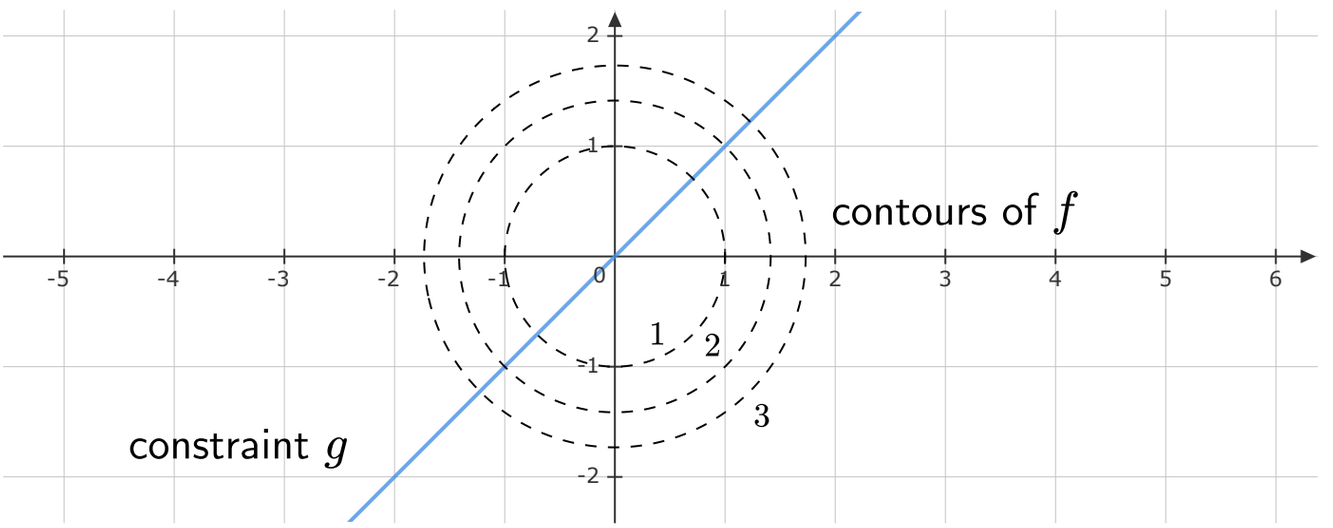
8.1. Example-1

$$\min_{x,y} \quad x^2 + y^2$$

$$y - x = 0$$

$$f(x, y) = x^2 + y^2$$
$$g(x, y) = y - x$$

The dotted curves are the contours of f. Recall that a contour is a curve where the function's value is the same at all points on the curve.

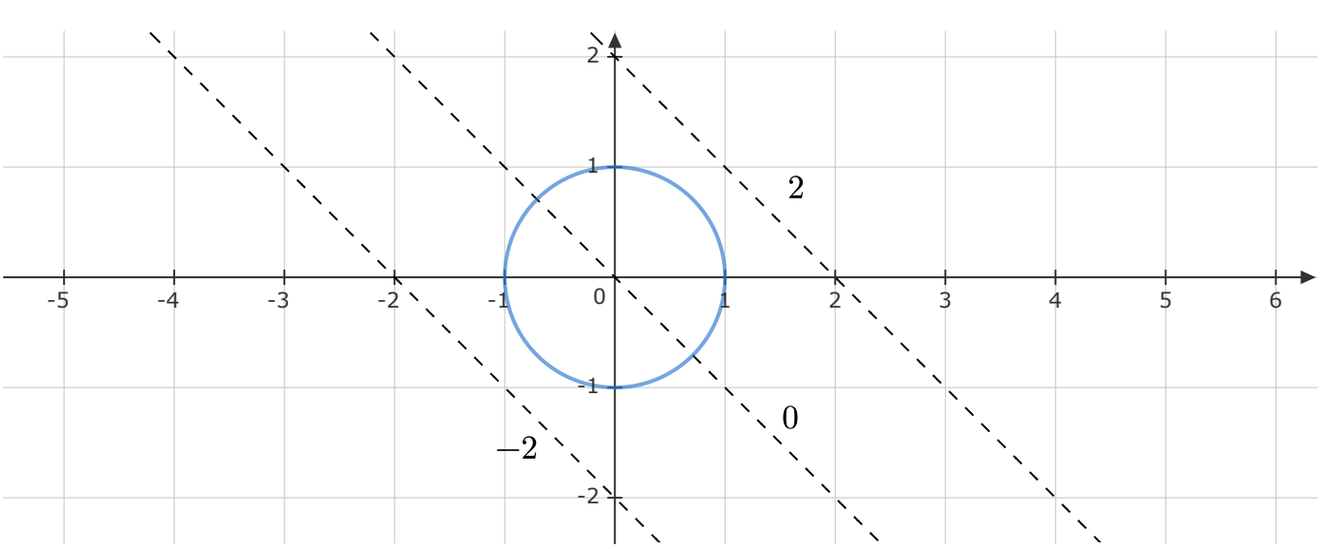


8.2. Example-2

$$\min_{x,y} \quad x + y$$

$$x^2 + y^2 - 1 = 0$$

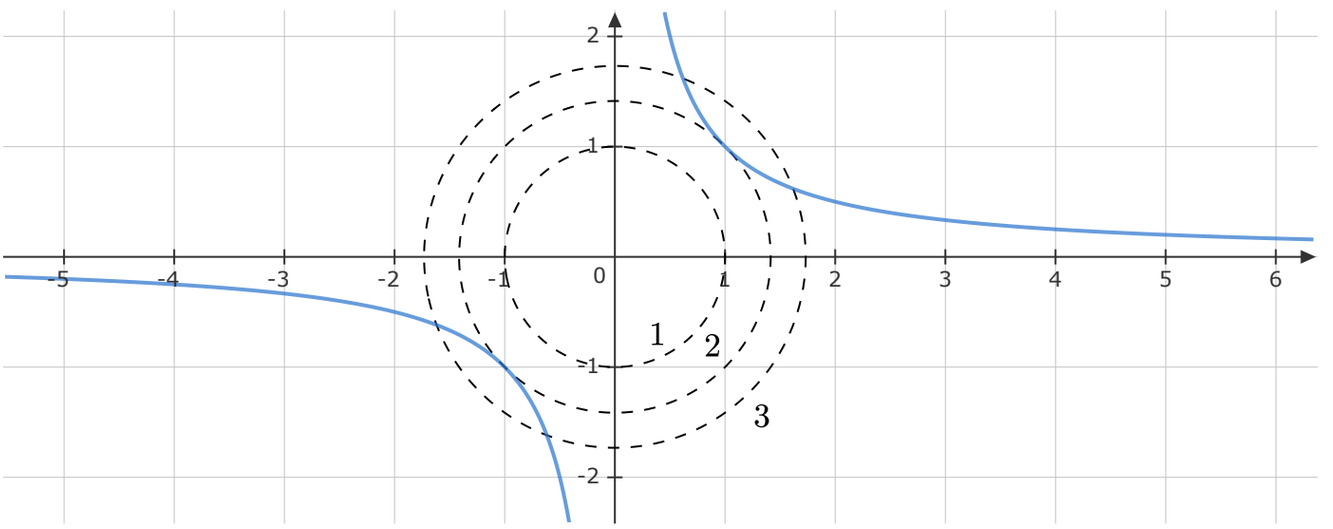
If you slice one plane with another plane, you get lines.



8.3. Example-3

$$\min_{x,y} \quad x^2 + y^2$$

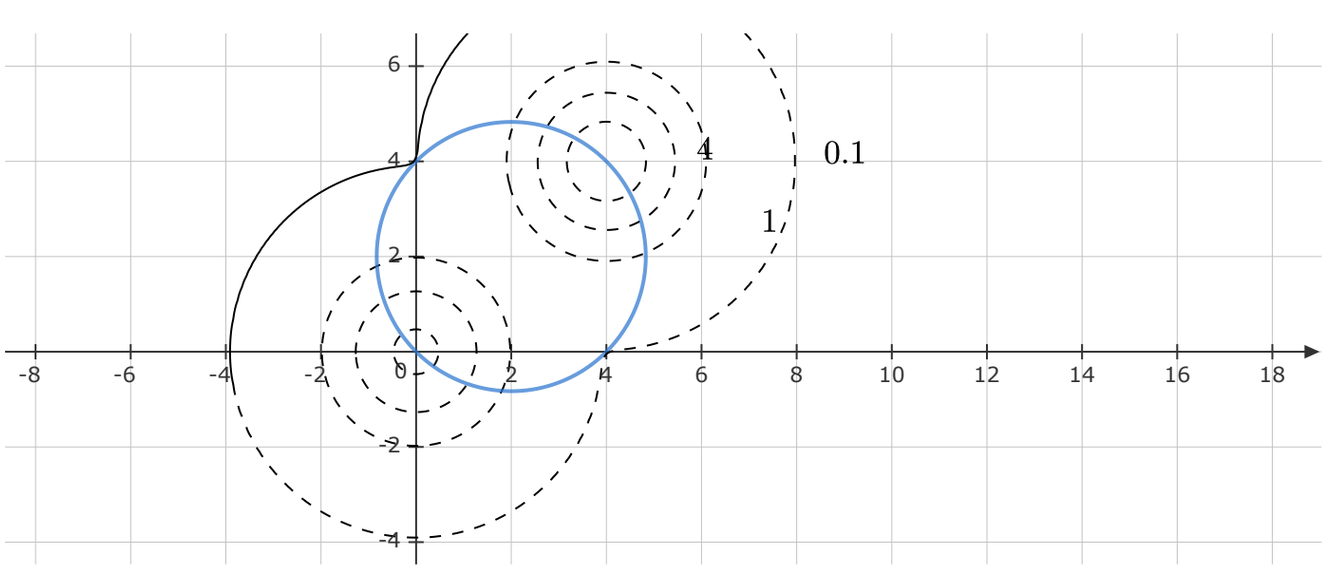
$$xy - 1 = 0$$



8.4. Example-4

$$\min_{x,y} \quad 5e^{-\left(x^2+y^2\right)} + 8e^{-\left[(x-4)^2+(y-4)^2\right]}$$

$$(x-2)^2 + (y-2)^2 - 8 = 0$$



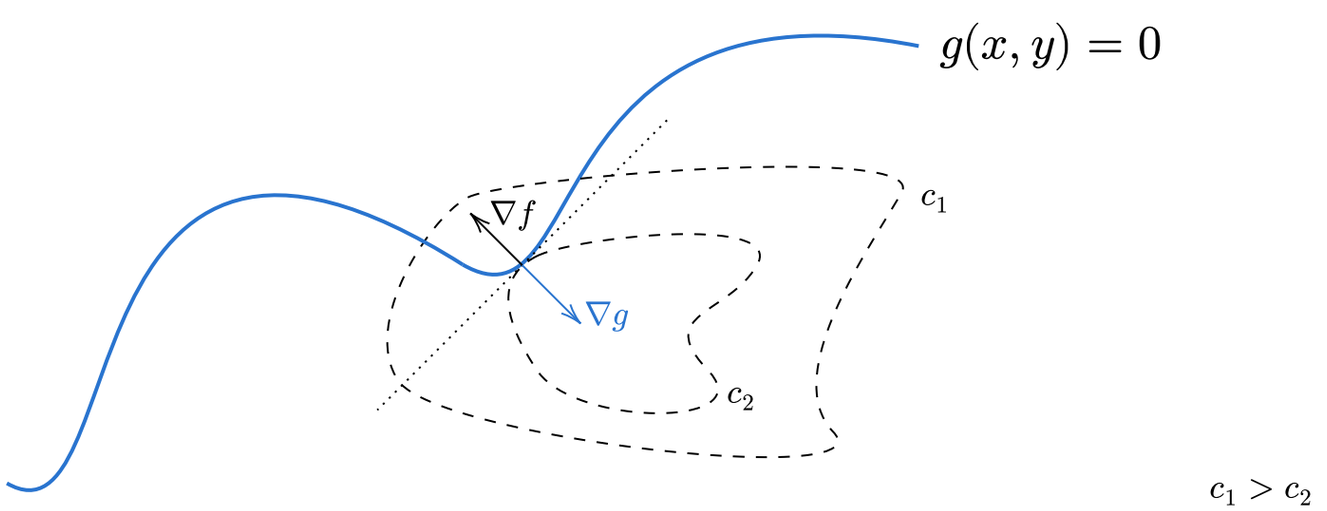
8.5. Necessary Conditions for Local Minimum

Back to the general form:

$$\min_{x,y} \; f(x,y)$$

$$g(x,y) = 0$$

If $\left(x^*, y^*\right)$ is a local minimum of f subject to g, then what can we say about ∇f and ∇g at that point? They have to be parallel. We will see why this has to be the case by first understanding descent directions and feasible directions.



If x^* is a minimum and we move by a small amount away from x^*, then what should ideally happen? As long as we satisfy the equality constraint, the function's value should not decrease. So we see that both the objective function and the constraint are involved. This brings up two directions of interest:

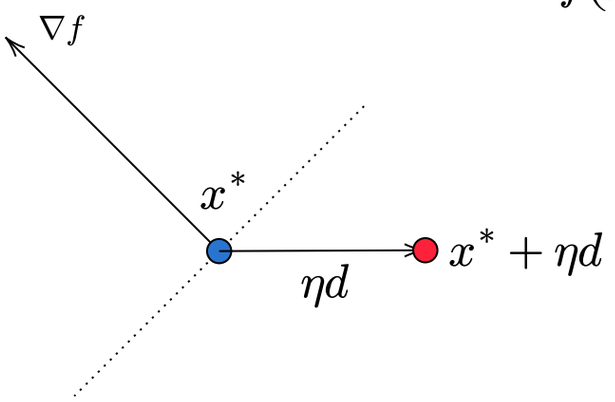
- descent direction: a direction along which the objective function's value decreases
- feasible direction: a direction along which the constraint is still satisfied

Descent directions

Let us first look at descent directions at a point x^*.

$$f(x^* + \eta d) < f(x^*)$$

zoomed-in image



Using Taylor series:

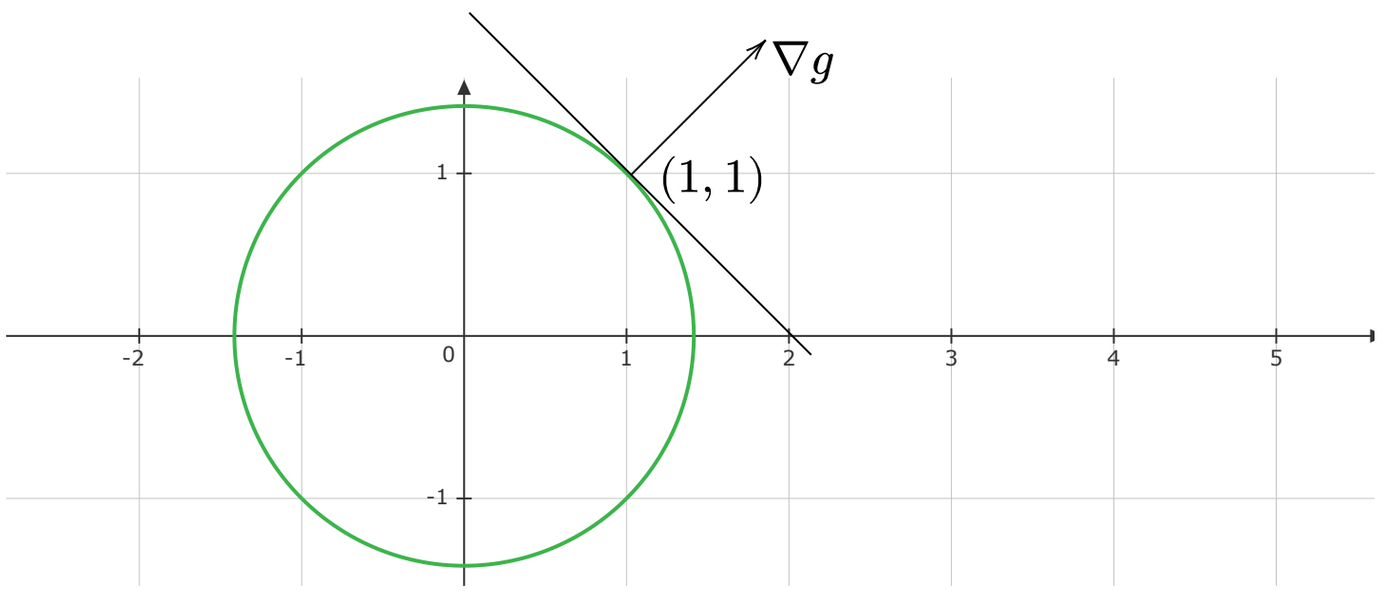
$$\begin{aligned} f(x^* + \eta d) &\approx f(x^*) + \nabla f(x^*)^T (\eta d) \\ &< f(x^*) \end{aligned}$$

The set of descent directions are therefore the ones where $\nabla f(x^*)^T (\eta d)$ is negative. In other words, all those directions that make an obtuse angle with ∇f.

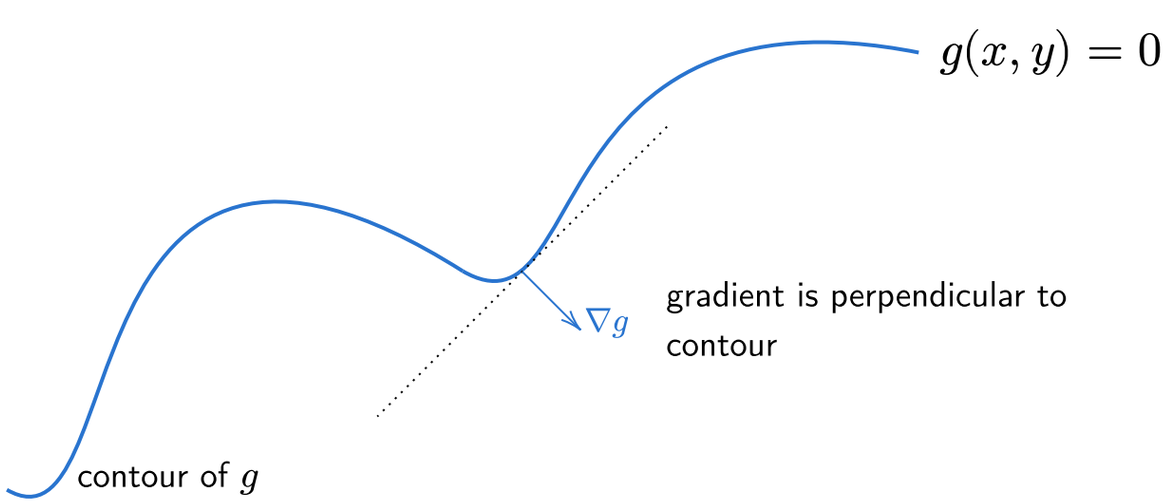
Feasible directions for equality constraints

Before we get there, let us quickly revisit the relationship between contours and gradients.

Gradient and Contours: The gradient to a function at a point is perpendicular to its contour at that point. The following is a contour of $g(x, y) = x^2 + y^2$. What is the gradient to g at $(1, 1)$? $\nabla g = (2, 2)$



Now, consider a curve representing the equality constraint.



What are the directions at x^* along which the constraint remains feasible? Going back to the Taylor series expansion around x^*:

$$g(x^* + \eta d) \approx g(x^*) + \nabla g(x^*)^T (\eta d)$$

If the constraint is satisfied at x^*, then $g(x^*) = 0$. If we want the constraint to be satisfied even after moving to $x^* + \eta d$, then $g(x^* + \eta d) = 0$. Therefore, up to first order, we have:

$$0 \approx 0 + \nabla g(x^*)^T (\eta d) \implies d \perp \nabla g(x^*)$$

We see that the only direction that remains feasible is perpendicular to the gradient, which is along the tangent to the contour at that point.

Descent and Feasible Directions

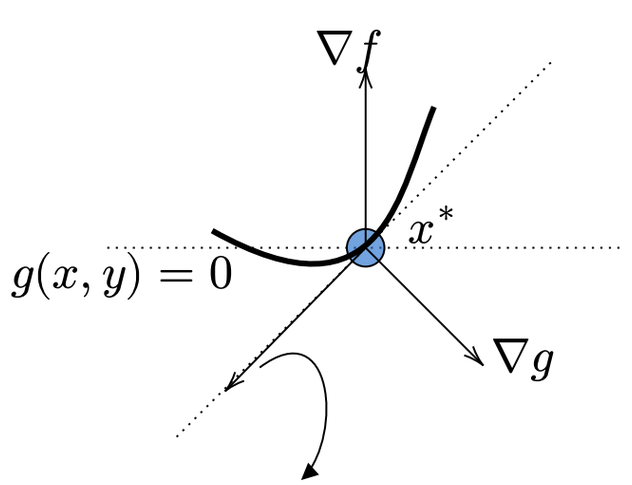
We are now ready to tackle the relationship between these two directions.

$$\min \;\; f(x, y)$$

$$g(x, y) = 0$$

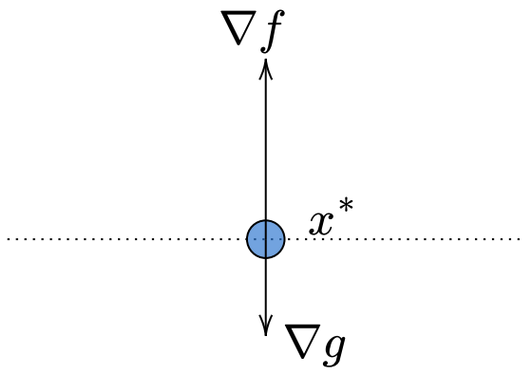
Consider a situation when ∇f and ∇g are not parallel. We then see that there is a descent direction that is feasible. This means that x^* cannot be a minimum as claimed.

$$||d|| \ll 1$$

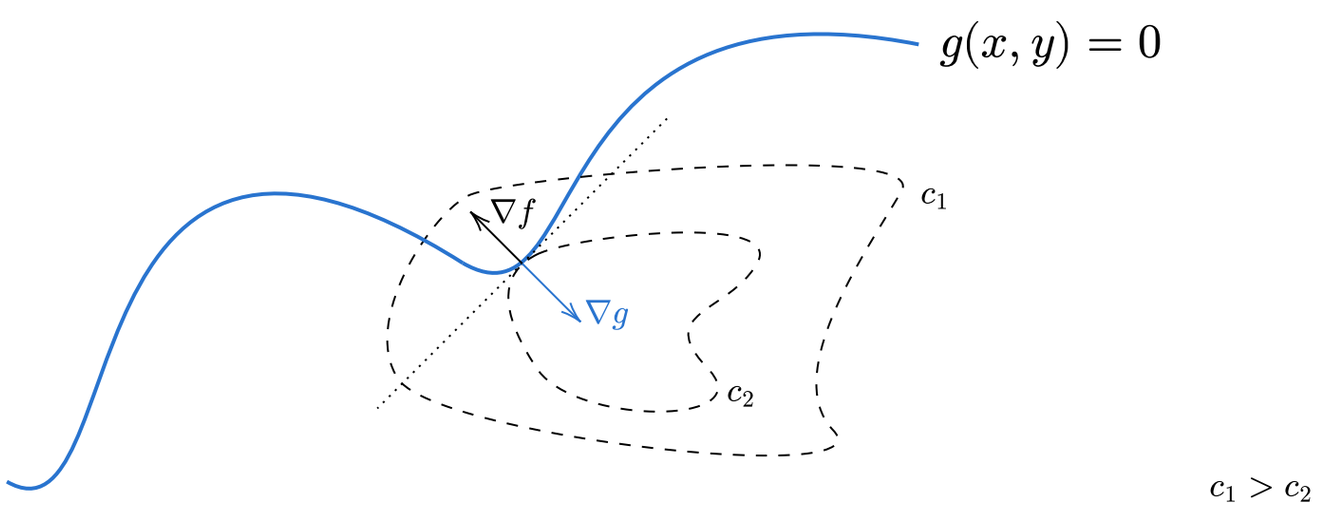


Descent direction that is still feasible $\implies x^*$ is not a minimum

Therefore, we are left with the conclusion that ∇f and ∇g are parallel.



The only way in which a descent direction is not feasible is when $\nabla f = \lambda \nabla g$ for some $\lambda \in \mathbb{R}$. They could be parallel or anti-parallel. Back to the original diagram we started with. Note that the contour of the constraint is perpendicular to the contour of the objective function at the point of minimum.



We have done all of this assuming that $\nabla g(x^*) \neq 0$. This is what we call a constraint qualification. We now present the following important theorem.

Theorem: If x^* is a minimum of the constrained optimization problem $\begin{array}{l} \min\limits_{x} \ f(x) \\ g(x) = 0 \end{array}$ and $\nabla g(x^*) \neq 0$ [constraint qualification] then there exists some $\lambda \in \mathbb{R}$ such that $\nabla f(x^*) + \lambda \nabla g(x^*) = 0$.

We can cast it in the form $P \implies Q$, where:

- P: x^* is a minimum and $\nabla g(x^*) \neq 0$ [constraint qualification]

- Q: There is some $\lambda \in \mathbb{R}$ such that $\nabla f(x^*) + \lambda \nabla g(x^*) = 0$

Compare this with the unconstrained case, where we said that if x^* is a minimum, then $\nabla f(x^*) = 0$. We can make a similar statement here, but with a new function called the Lagrangian.

Lagrangian: The Lagrangian function for a constrained optimization $\begin{array}{l} \min\limits_{x,y} \ f(x,y) \\ g(x,y) = 0 \end{array}$ problem is $L(x,y,\lambda) = f(x,y) + \lambda g(x,y)$.

The necessary condition for a minimum can be reformulated as follows:

Theorem: If x^* is a minimum of the constrained optimization problem $\begin{array}{l} \min\limits_{x} \ f(x) \\ g(x) = 0 \end{array}$ and $\nabla g(x^*) \neq 0$ [constraint qualification] then there exists some $\lambda \in \mathbb{R}$ such that $\nabla L(x^*, \lambda) = 0$.

9. Method of Lagrange Multiplier

The method of Lagrange multipliers makes use of the above theorem to narrow down the potential minima for a constrained optimization problem with equality constraints. Let us look at a few examples.

9.1. Example-1

$$\min_{x,y} \quad x + y$$

$$x^2 + y^2 - 1 = 0$$

Step-1: Define the function and the constraint

$$\begin{aligned} f(x,y) &= x + y \\ g(x,y) &= x^2 + y^2 - 1 \end{aligned}$$

Step-2: Define the Lagrangian function

$$L(x, y, \lambda) = f(x, y) + \lambda g(x, y)$$

$$L(x, y, \lambda) = x + y + \lambda\left(x^2 + y^2 - 1\right)$$

Step-3: Set $\nabla L = 0$

$$\begin{aligned} 1 + 2\lambda x &= 0 \\ 1 + 2\lambda y &= 0 \\ x^2 + y^2 - 1 &= 0 \end{aligned}$$

Step-4: Solve for x, y, λ

$$x = y = \frac{-1}{2\lambda}$$

Substitute this into the third equation from the previous step:

$$\frac{2}{4\lambda^2} - 1 = 0 \Longrightarrow \lambda = \pm\frac{1}{\sqrt{2}}$$

Therefore, the candidates for minima are:

$$(x_1, y_1) = \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$$

$$(x_2, y_2) = \left(\frac{-1}{\sqrt{2}}, \frac{-1}{\sqrt{2}}\right)$$

Geometrically, the objective function represents a plane in $\mathbb{R}^3$. The constraint represents a hollow cylinder in $\mathbb{R}^3$. The minimum in this case is the lowest point on the plane that intersects the cylinder. This kind of geometric interpretation may not always be possible.

9.2. Example-2

Find the points on the circle $x^2 + y^2 = 80$ which are closest and farthest from the point $(1, 2)$.

Step-1: Find out the objective and the constraint

$$\begin{aligned} f(x, y) &= (x-1)^2 + (y-2)^2 \\ g(x, y) &= x^2 + y^2 - 80 \end{aligned}$$

Step-2: Find the Lagrangian, L

$$L(x, y, \lambda) = (x-1)^2 + (y-2)^2 + \lambda\big(x^2 + y^2 - 80\big)$$

Step-3: $\nabla L = 0$

$$\begin{aligned} 2(x-1) + 2\lambda x &= 0 \\ 2(y-2) + 2\lambda y &= 0 \\ x^2 + y^2 &= 80 \end{aligned}$$

Step-4: Solve for x, y, λ

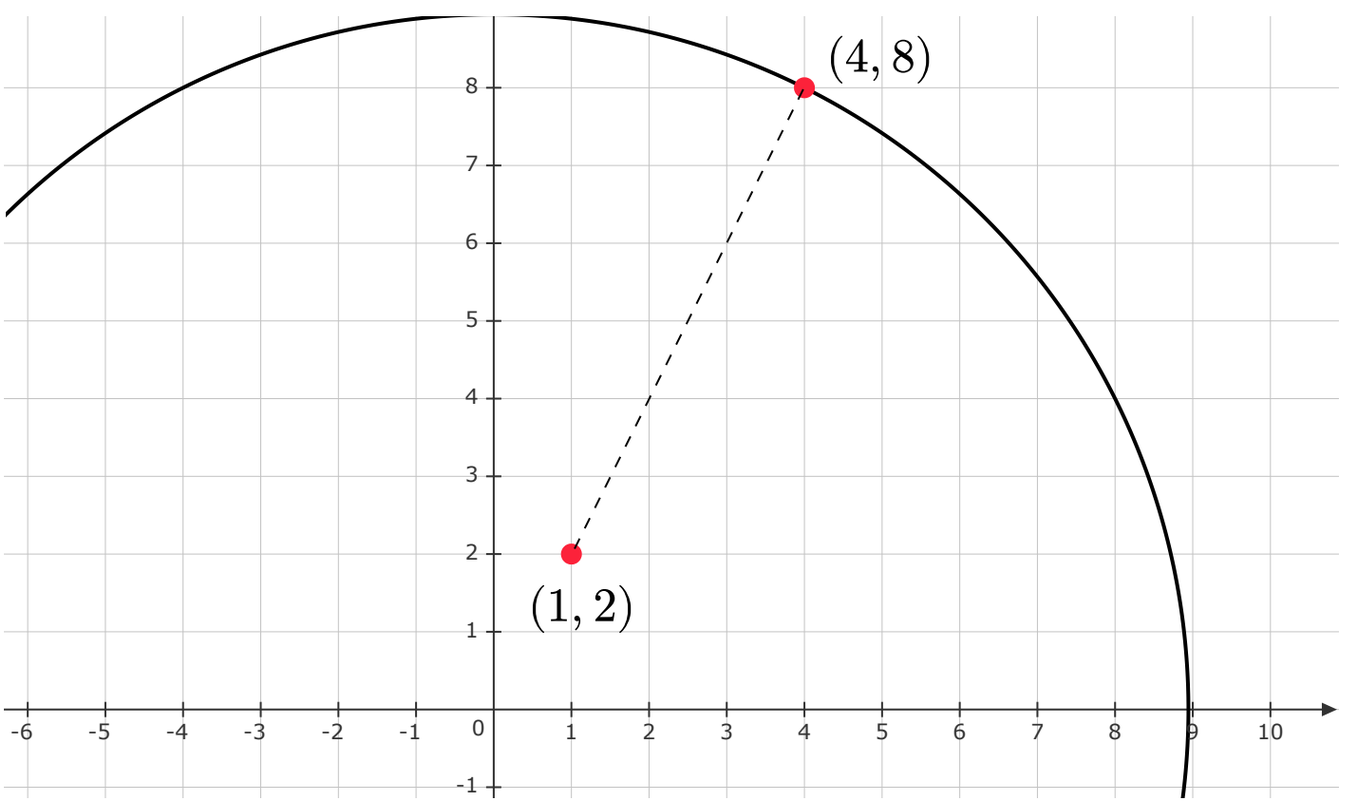
Observe that $x, y, \lambda \neq 0$. Also $x \neq 1, y \neq 2$.

$$\frac{x-1}{y-2} = \frac{x}{y} \implies xy - y = xy - 2x \implies y = 2x$$

Plugging this into the constraint:

$$x^2 + 4x^2 = 80 \Longrightarrow x^2 = 16 \Longrightarrow x = \pm 4$$

Therefore, we have the following candidates:

$$(4, 8), \ (-4, -8)$$

Once again, we can try to get a geometric interpretation and show that one of these has to correspond to a minimum: